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Studierichting: Informatica
Oefeningenreeks 3G nr. 13.2

$$f(x) = \sin 3x$$

1. Domein

$$\text{Dom } f = \mathbb{R}$$

$$p = \sin 3x \Leftrightarrow 3x = 2\pi \Leftrightarrow x = \frac{2\pi}{3}$$

$$\Rightarrow \text{Beperkte periode : } \left[0, \frac{2\pi}{3}\right]$$

2. Asymptoten

er zijn geen asymptoten.

3. Tekenonderzoek van $f(x)$

$$\sin 3x = 0$$

$$3x = k\pi$$

$$x = \frac{k\pi}{3}$$

$$\Rightarrow f^{(-1)}(0) = \left\{0, \frac{\pi}{3}, \frac{2\pi}{3}\right\}$$

x	$\left \begin{array}{ccc} 0 & \frac{\pi}{3} & \frac{2\pi}{3} \end{array} \right.$
$f(x)$	$\left \begin{array}{ccc} 0 & + & 0 \\ 0 & - & 0 \end{array} \right.$

4. Tekenonderzoek van $f'(x)$

$$f(x) = \sin 3x$$

$$f'(x) = 3 \cos 3x$$

$$3 \cos 3x = 0$$

$$\cos 3x = 0$$

$$3x = \frac{\pi}{2} + k\pi$$

$$x = \frac{\pi}{6} + \frac{k\pi}{3}$$

$$\Rightarrow f'^{(-1)}(0) = \left\{ \frac{\pi}{6}, \frac{\pi}{2} \right\}$$

x	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$
$f'(x)$	$+$	0	$-$

$$M\left(\frac{\pi}{6}\right) = \sin\left(3 \cdot \frac{\pi}{6}\right)$$

$$M\left(\frac{\pi}{6}\right) = 1$$

$$\Rightarrow M\left(\frac{\pi}{6}, 1\right)$$

$$m\left(\frac{\pi}{2}\right) = \sin\left(3 \cdot \frac{\pi}{2}\right)$$

$$m\left(\frac{\pi}{2}\right) = -1$$

$$\Rightarrow m\left(\frac{\pi}{2}, -1\right)$$

5. Tekenonderzoek van $f''(x)$

$$f'(x) = 3 \cos 3x$$

$$f''(x) = -9 \sin 3x$$

$$-9 \sin 3x = 0$$

$$\sin 3x = 0$$

$$3x = k\pi$$

$$x = \frac{k\pi}{3}$$

$$\Rightarrow f''^{(-1)}(0) = \left\{ 0, \frac{\pi}{3}, \frac{2\pi}{3} \right\}$$

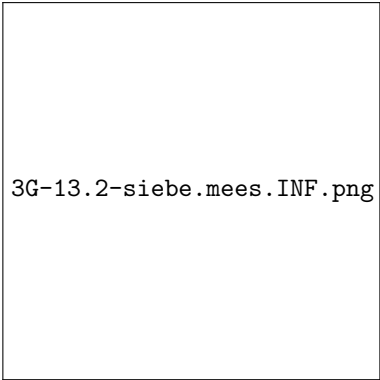
x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$
$f''(x)$	0	$-$	$+$

$$B\left(\frac{\pi}{3}\right) = \sin\left(3 \cdot \frac{\pi}{3}\right)$$

$$B\left(\frac{\pi}{3}\right)=0$$

$$\Rightarrow B\left(\frac{\pi}{3},0\right)$$

6. Schets



⇒ Samenvattende tabel

x	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$
$f(x)$	0	+	0	−	0
$f'(x)$	+	0	−	0	+
$f''(x)$	0	+	0	−	0
$f(x)$	$\nearrow$	$M(1)$ $\smile$	$\searrow$ $B(0)$	$m(-1)$ $\frown$	$\nearrow$

References